

Q Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$

Sol Lagrange's auxiliary eqn is

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$$

$$\Rightarrow \frac{dy}{2xy} = \frac{dz}{2xz}$$

$$\Rightarrow \log y = \log z + \log c$$

$$\Rightarrow y = cz \Rightarrow c = \frac{y}{z} \quad \text{--- (1)}$$

Now taking multipliers x, y & z

$$\frac{x dx + y dy + z dz}{x^3 - xy^2 - xz^2 + 2xy^2 + 2xz^2} = \frac{dx}{2xz}$$

$$\Rightarrow \frac{x dx + y dy + z dz}{x(x^2 + y^2 + z^2)} = \frac{dx}{2xz}$$

$$\Rightarrow \frac{2(x dx + y dy + z dz)}{2x(x^2 + y^2 + z^2)} = \frac{dx}{2xz}$$

on integrating

$$\int \frac{2(x dx + y dy + z dz)}{x^2 + y^2 + z^2} = \int \frac{dx}{xz}$$

$$\log(x^2 + y^2 + z^2) = \log z + \log c_2$$

$$\frac{x^2 + y^2 + z^2}{z} = c_2$$

hence the complete solⁿ is

$$f\left(\frac{x^2 + y^2 + z^2}{z}, \frac{y}{z}\right) = 0.$$

③ Solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

4) Solve
 $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2).$

5) Solve

$$(mx - ny) \frac{\partial z}{\partial x} + (nx - lz) \frac{\partial z}{\partial y} = ly - mx$$

Sol here $P = mx - ny$, $Q = nx - lz$, $R = ly - mx$

Lagrange's auxiliary eqⁿ is

$$\frac{dx}{mx - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}$$

→ taking x, y, z as multipliers

$$\frac{x dx + y dy + z dz}{\cancel{mxz} - \cancel{nyy} + \cancel{nyy} - \cancel{lzy} + \cancel{lyz} - \cancel{mxx}} = 0$$

$$\text{So } x dx + y dy + z dz = 0.$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C$$

$$\Rightarrow x^2 + y^2 + z^2 = C_1 \quad \text{--- (1)}$$

then using l, m, n as multipliers

$$\frac{l dx + m dy + n dz}{\cancel{l x z} - \cancel{l y y} + \cancel{n x m} - \cancel{l z m} + \cancel{l y n} - \cancel{m x n}}$$

$$\text{So } l dx + m dy + n dz = 0$$

$$\text{on integrating, } lx + my + nz = C_2$$

thus the complete soln is

$$\Phi(x^2 + y^2 + z^2, lx + my + nz) = 0 \text{ Ans.}$$